

Extremum problems

Lessons 23–25 · Algebra 11, §1.8 · P1 · 7.8 · 28 problems

Name the variable, write the quantity to be optimised in terms of that one variable, then differentiate. Use the constraint to remove the second variable **before** differentiating, and state the units in the answer.

NAME _____

CLASS _____

DATE _____

Easy 7

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| 1 Two numbers add to 30. Maximise their product. | 2 Two numbers add to 10. Minimise the sum of their squares. |
| 3 Maximum value of $y = 12x - x^2$ | 4 Minimum value of $y = x^2 - 8x + 3$ |
| 5 200 m of fence makes a rectangular field. Find the greatest area. | 6 A square of side x is cut from each corner of a 16×16 card. Give the domain of x . |
| 7 A rectangle has perimeter 24. Write its area in terms of one side x . | |

Medium 7

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| 8 60 m of fence encloses a rectangle against a wall (three sides). Find the greatest area. | 9 Corner squares of side x are cut from a 24×24 card and the sides folded up. Find the greatest volume. |
| 10 A closed cylinder holds 2000 cm^3 . Find the radius that minimises the surface area. | 11 Least value of $y = x + \frac{16}{x}$ for $x > 0$ |
| 12 Greatest value of $y = x^3 - 6x^2$ on $[0, 5]$ | 13 A rectangle has area 64 cm^2 . Minimise its perimeter. |
| 14 An open box has a square base of side x and volume 500 cm^3 . Write its surface area in terms of x . | |

Hard 7

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| 15 A rectangle is inscribed in a circle of radius 10. Maximise its perimeter. | |
| 16 Find the point on $y = 2x + 1$ closest to the origin. | |
| 17 A cylinder is inscribed in a sphere of radius 3. Maximise its volume. | |
| 18 A wire 100 cm long is cut into two pieces, one bent into a square and the other into an equilateral triangle. Minimise the total area. | |
| 19 An open-topped box with a square base holds 4000 cm^3 . Minimise the material used. | |
| 20 A running track is a rectangle with a semicircular end at each side, of total perimeter 400 m. Maximise the area of the rectangle . | |
| 21 A box with a square base and no lid is made from 300 cm^2 of card. Maximise its volume. | |

Very hard 7

- 22 Show that among all rectangles of a given perimeter, the square has the greatest area.
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- 23 A cylinder is inscribed in a cone of height 12 and base radius 5, with its base on the base of the cone. Maximise its volume.
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- 24 A pipeline runs from a rig 5 km offshore to a refinery 13 km along a straight coast. Underwater pipe costs \$5m per km and land pipe \$3m per km. Where should the pipe meet the shore, and what is the least cost?
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- 25 The height plus the circumference of a cylindrical parcel may not exceed 300 cm. Maximise its volume.
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- 26 Show that the rectangle of greatest area inscribed in a semicircle of radius r , with one side on the diameter, has area r^2 .
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- 27 Find the point on $y = x^2$ closest to $(3, 0)$.
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- 28 A cylindrical can with no lid has volume V . Show that the material is least when the height equals the radius.
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